The Effect of the Bologna Reform on the Graduate Labor Market *

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Abstract

In 2011, the Bologna reform shortened the duration of undergraduate programs by one year and reduced the content of the curricula in Russia. Notably, it did not change the quantity nor the quality of admitted students. Using this reform as a natural experiment, I find no adverse effect of a 20% reduction in the length of higher education on employment. There are also no adverse effects on wages for graduates who worked less during their studies. I provide suggestive evidence showing that students could compensate for the loss of their human capital by investing more time studying.

Keywords: difference–in–differences, event–study framework, higher education, human capital, returns to education, Russia

JEL Codes: I23, I26, J24

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1 Introduction

The Bologna reform is the largest education reform of the 21st century in Europe. The main goal was to improve higher education participation, study success, and labor market outcomes of graduates (Teichler, 2011). The core step to implement the reform was to create a two–cycle degree structure: bachelor's and master's. Subsequently, the length of university programs has been reduced. This raises a question: has the Bologna reform negatively affected the labor market outcomes of graduates as predicted by the human capital theory?

In Russia, the Bologna reform transformed higher education programs from 5 to 4 years, while some programs were not affected by the reform. Unlike in other countries which implemented the Bologna reform, students in Russian universities graduate within the nominal study period. They rarely take longer than their official study duration due to administrative barriers and financial costs. This provides a credible identification strategy to estimate the effect of the Bologna reform. Besides, universities and policymakers have been questioning the benefits of the Bologna reform and advocating for the reverse of the process to the Soviet–style unitary higher education system. Hence, providing credible estimates of the impact of the Bologna reform is essential for policymakers to base their wide–ranging decisions on empirical evidence.

This paper estimates the effect of the Bologna reform on wages and employment in Russia. The main impact it had on higher education is the reduction in the length of university programs by one year. Hence, the reform allows me to use a difference–in–differences design to estimate the causal effect of one year of higher education. The identifying assumption is that, in the absence of the Bologna reform, the quantity and the quality of university applicants and graduates would have evolved similarly in programs that experienced a one–year reduction of education relative to programs that did not experience such a change.

The results show that there are no detrimental effects of the Bologna reform on the employment of graduates (the effect is –0.014, s.e. = 0.007). I show that the OLS estimates of log monthly wages yield returns of about 5% to one year of higher education. These estimates are similar to the results found in the literature about the Russian labor market (Belskaya et al., 2020; Kyui, 2016). However, my findings

show that returns to one year of higher education are not statistically different from zero. This suggests that the OLS results are biased upwards since people with higher marginal returns to education tend to choose higher levels of schooling. Notably, there are no effects of the reform on wages of female graduates (the effect is 0.017, s.e. = 0.015) and significant adverse effects for males (the effect is –0.046, s.e. = 0.015).

A possible mechanism of no adverse effects of the Bologna reform on employment and female wages is that students could compensate for the reduction of their human capital. Arteaga (2018) shows that a reduction in the length of higher education in Columbia resulted in a reduction in the number of courses which decreased students' performance in the recruitment process and ultimately decreased their wages. As a result, the author rejects a pure signaling model. In line with Arteaga (2018), I support the human capital theory and argue that the heterogeneous results of the reform on wages are due to different strategies chosen by students. The reform did not lead to a reduction in the accumulation of human capital of students who could compensate for a one–year reduction by working less during their studies and studying more. To support this claim, I offer suggestive evidence and show that female students worked less during their studies (the effect is –0.042, s.e. = 0.011), while males chose to work more (the effect is 0.081, s.e. = 0.015). Subsequently, female graduates were less likely to get on-the-job training due to insufficient competence (the effect is –0.045, s.e. = 0.013), whereas males had to get more training (the effect is 0.072, s.e. = 0.024). As a result, female students invested more in the accumulation of their human capital which allowed them to compensate for the reduction in the study duration.

This paper offers two important contributions to the literature. First, it enriches the literature focusing on evaluating the results of the Bologna reform in European countries. This paper is the first to study the impact of the Bologna reform on the labor market outcomes in Russia. The implementation of the reform had mixed effects on the labor market outcomes of graduates in European countries. Several papers find a significant negative effect of the Bologna reform (Neugebauera and Weiss, 2018; Farčnik and Domadenik, 2012), while others find no impact of the reform (Garra, 2013; Bosio and Leonardi, 2011).

Second, it contributes to the emerging literature on the causal estimates of eco-

nomic returns to higher education in middle–income countries (Dai et al., 2021; Arteaga, 2018). The closest study to mine is Arteaga (2018), which studies a reform at Colombia's top university, but the current paper differs in two important ways. I focus on a national reform that affects all universities, while Arteaga (2018) studies a reform at one university. Besides, the Bologna reform affected students in all fields of study in Russia, whereas Arteaga (2018) estimates an effect of the reform only among graduates in economics and business departments.

I consider three possible threats to the identification strategy, e.g. changes in the cohort composition, an introduction of the standardized exam for admission to universities, and the 2015 Russian financial crisis. I show that it is unlikely that university applicants could affect the year of their application as about 90% of school graduates apply in the same year as they graduate from school. Furthermore, I check whether the reform changed the pool of admitted students and graduates. For instance, as earning a higher education degree is easier, lower–ability individuals could be motivated to enroll in 4–year programs. I show that the reform did not change the quantity nor the quality of admitted students and graduates. Moreover, I show that the characteristics of students did not change due to the introduction of the standardized exam for admission to universities in 2009. Lastly, in 2015, there was a financial crisis in Russia, which could have influenced labor market conditions for graduates. To address this concern, I use the event-study framework and include students, who graduated in 2010–2018, to estimate the effect of one year of higher education on employment and find no impact of the reform over a longer period.

The paper is structured as follows. Section 2 reviews the institutional background of the Russian higher education system and the Bologna reform. Section 3 presents the data and describes the identification strategy. I present the results and robustness checks and discuss a possible mechanism in Section 4. Section 5 concludes.

2 Institutional background and the reform

The Russian higher education system

All students sit the standardized exam for admission to universities – the Unified State Exam (similar to the SAT) – in their last grade at high school. The higher the score is, the more chances a student gets admitted to a preferred university. Traditionally, about 90% of university applicants are students who apply to university in the same year as they graduate from school.

The Ministry of Science and Higher Education determines the number of state–funded places and the curricula in universities.[1] The majority of students have state–funded places in higher education institutions in Russia. Some universities, which are under the supervision of sectoral ministries, such as the Ministry of Agriculture, the Ministry of Culture, and the Ministry of Health, set their regulations for universities, although they have little power to influence the curricula and the quality of programs.

The Bologna reform in Russia

The two–cycle degree structure has been optional for universities, parallel to 5–year degrees, since 1992. In 2003, Russia signed the Bologna Declaration and some universities started implementing two–cycle degree programs. In 2011, universities, which were under the supervision of the Ministry of Science and Higher Education and several others, transformed all undergraduate programs from 5– to 4–year degrees, while other ministries did not require universities to follow this reform.[2] Thus, if applicants applied to a specific program, they could only apply to either 4– or 5–year programs. Such an institutional design allows me to use those programs that were transformed from 5 to 4 years as a treatment group, and those that remained as 5 years as a control group. Importantly, there was no change in the high school

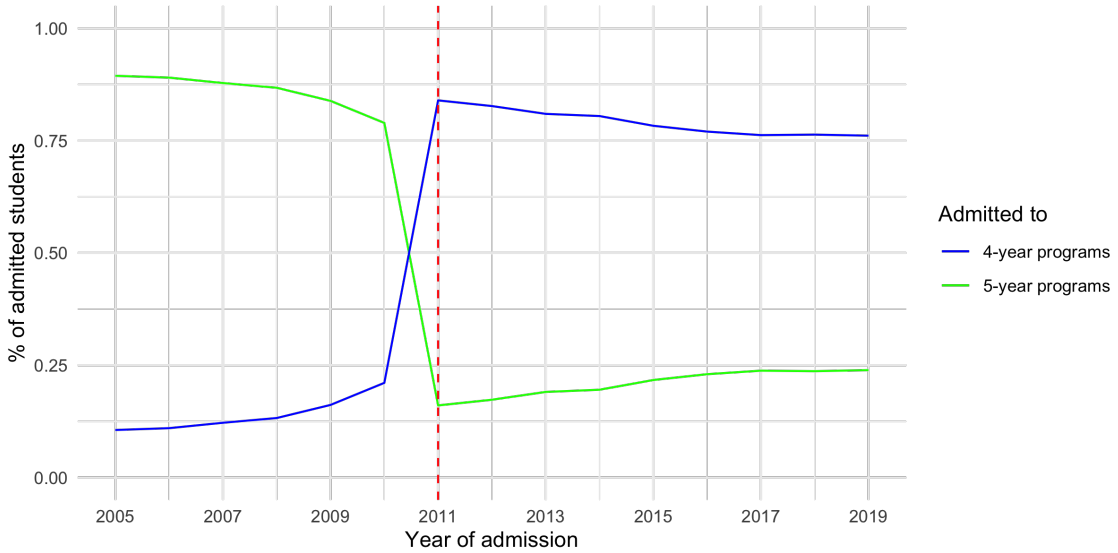
[1] In this paper I describe the system of higher education that is relevant for full–time students in public and private universities. Hereinafter by universities are meant all types of higher education institutions, including universities, institutes, and academies.

[2] The ministries that did not require universities to reduce the length of programs followed a Decree of the Government of the Russian Federation N1136, which specified certain programs within broad fields of study to keep at 5 years. It was a politically–oriented decision to keep programs at 5 years and it did not relate to labor market conditions.

curricula to adjust for the reduction in the university study period, so university applicants are homogeneous over time (see Section 4.4 for more details).

The Bologna reform transformed Russian higher education from a unitary system, with 5–year specialists' degrees, to the two–cycle system, with 4–year bachelor's and 2–year master's programs. In general, some parts of the course structure and curricula were reduced, especially optional elements. A one–year reduction of the curricula was mostly accompanied by the reduction of specialized courses, so the so–called "specialization" of higher education was reduced. This is one of the most fundamental institutional reforms of the Russian higher education system in the last decade and one of the most debated ones. The mass introduction of two–cycle higher education degrees happened in the academic year 2010–2011. In 2010, 79% students were admitted to 5–year programs and 21% to 4–year programs and the opposite pattern in 2011: 16% were 5-year students and 84% were 4-year students (see Figure 1). This creates a natural experiment for the 2011 intake.

Figure 1: The share of 4– and 5–year full–time admitted students in public and private universities in 2005–2019



Notes: The red dashed line shows the year of the implementation of the reform.
Source: Ministry of Science and Higher Education.

Two main reasons make the case of Russia particularly interesting. First, the reform was exogenously implemented in the university sector in 2011, thus, univer-

sities and students could not affect their decision to participate in the reform or not. I consider the reform as a source of an exogenous variation in the length of the university study period. This reform generates the exogenous assignment of students into the 4–year treatment or 5–year control groups. Since the year of application to the university is a function of the year of graduation from school, this policy change provides exogenous variation in the length of the university study period. Second, the duration of programs is strictly limited to the nominal study period. In other words, students have to graduate within the study limit period due to administrative barriers and financial costs. Thus, I assume that any changes in the differences between the treated and control groups can be attributed to the effect of one year of higher education.

Roshchin and Rudakov (2015) and Lukiyanova (2010) summarise a wide variety of papers based on Russian labor market data, which are dedicated to estimating returns to education. All in all, the majority of the studies for Russia rely on OLS estimates. Belskaya et al. (2020) show that returns to one year of higher education are 7.3% in Russia. Kyui (2016) identifies strong positive returns to higher education in terms of employment and wages. Ultimately, there are no causal estimates of how the Bologna reform affected the graduate labor market in Russia, particularly wages and employment.

3 Empirical approach

3.1 Data

The data are obtained from the National Survey of Graduate Employment carried out by the Russian Federal State Statistics Service (Rosstat). This is a repeated cross–sectional sample of students who graduated in 2010–2015. The sample is representative of the national level. The survey provides extensive information on educational backgrounds and labor market experiences during the first years after graduation.

I restrict data to 4– and 5–year university graduates. The education variable refers to the highest level of education attained.[3] According to the definition of

[3]When respondents were asked to specify their higher education degree, they could only choose

youth by Rosstat, I further restrict the dataset to 20–30 years old graduates who are not currently students. Only graduates from full–time programs are included. In total, there are 11,332 observations.

I use net monthly wages and net hourly wages, calculated from net monthly wages, the number of working days, and working hours per day.[4] The unemployed category captures both the registered unemployed and the economically inactive.

Table 1 shows the descriptive statistics of the variables for 4– and 5–year university graduates, respectively. 41% and 42% of graduates are males. 94% and 95% of graduates study at public universities. 53% and 60% of students have state–funded places. 82% and 86% of graduates are employed in the year of the survey. The mean monthly wages in 2016 are 23,279 and 24,539 RUB (360 and 380 USD, 1 USD = 65 RUB).

Table 2 shows that 99–100% of 4–year and 92–97% of 5–year students graduate within the nominal study period which is essential for my identification strategy. Importantly, the dropout rate is low so the estimates of returns to education for the analyzed sample of graduates would not be biased.

3.2 Identification strategy

This paper estimates the short–term causal effect of one year of higher education on wages and employment in Russia. I do so by using the Bologna reform as a source of an exogenous variation in the length of higher education that is orthogonal to the potential outcomes. Using two cohorts of students who graduated in the same year, I can estimate the differences between the level of wages and the probability of being employed of 4–and 5–year degree holders under similar labor market conditions. However, the simple estimates of returns to higher education for these two groups

between "bachelor's" or "specialist's/master's". Although it may potentially create a bias in the estimates of the wages and the employment of the group "specialist's/master's", the share of masters' graduates in 2010–2015 is negligible. The share of master's graduates is about 4–8% out of all graduates in 2010–2015. Furthermore, I find it most plausible that the inclusion of master's graduates would bias the estimates towards exaggerating the returns to higher education. Hence, it would most likely overestimate the effect of a one–year reduction in higher education on wages and employment, and such an upward bias would only strengthen the null results found in this study.

[4] A third of employed university graduates did not report their wages. There are no statistically significant differences between university graduates who did and did not report their wages.

Table 1: Summary statistics

Variable	Observations	Mean	SD	Min	Max
4–year university graduates					
Male	1,722	0.41	0.49	0	1
Public institution	1,722	0.94	0.23	0	1
State–funded	1,722	0.53	0.50	0	1
Employed	1,722	0.82	0.38	0	1
Monthly wage, RUB	887	23,279	11,092	5,000	130,000
5–year university graduates					
Male	9,610	0.42	0.49	0	1
Public institution	9,610	0.95	0.21	0	1
State–funded	9,610	0.60	0.49	0	1
Employed	9,610	0.86	0.35	0	1
Monthly wage, RUB	5,320	24,539	12,009	5,000	120,000

Source: National Survey of Graduate Employment carried out by the Russian Federal State Statistics Service.

Table 2: The share of students who did not graduate within the nominal study period and the drop out rate

Variable	2013	2014	2015	2016	2017	2018
% of students who did not graduate within the nominal study period						
4–year programs	0.00	0.00	0.01	0.01	0.01	0.01
5–year programs	0.03	0.04	0.05	0.07	0.07	0.07
% of students who dropped out						
4–year programs	0.11	0.10	0.09	0.09	0.09	0.09
5–year programs	0.08	0.07	0.07	0.08	0.08	0.08

Source: Ministry of Science and Higher Education.

are confounded by the effect of other changes that had impacts on students enrolled before and after the reform. To correct for that I use a control group, i.e. graduates from programs that did not change in length. Thus, the changes in the differences between the treated and control groups under certain assumptions are explained by the effect of one year of higher education. A DiD design allows me to estimate the average treatment effect on the treated.[5]

The identifying assumption is that, in the absence of the Bologna reform, the quantity and the quality of university applicants and graduates would have evolved similarly in programs that experienced a one–year reduction of education relative to programs that did not experience such a change. In other words, I assume that graduates with a particular set of individual characteristics receiving the treatment would have, on average, experienced the same changes in their outcomes had they not received the treatment as individuals with the same set of individual characteristics who did not receive the treatment.

As described in the institutional background, most programs transformed from 5 to 4 years. The unit of treatment is a program, i.e. all students follow the nominal study duration of a specific program. Thus, the treatment group consists of graduates from programs that changed in length, the control group consists of graduates from programs that did not change in length. In 2010, almost all programs lasted for 5 years and the policy change occurred in 2011. As the year of application to the university is a function of school graduation year, university applicants could not affect their decision to start 4– or 5–year programs. In Section 4.4 I show that the Bologna reform indeed did not change the quantity nor the quality of admitted students. As a result, I argue that the Bologna reform offers an exogenous variation in the length of higher education. In other words, variation in the length of the university study period is assumed to be uncorrelated with other factors that might affect labor market outcomes, such as ability or self–selection.

[5] Although a DiD design is used to estimate the causal effect of a policy intervention when some individuals are subject to treatment and others not, and outcomes are measured in each group before and after the policy reform, individuals in the treated and control groups should not be necessarily the same individuals (Abadie et al., 2017; Athey and Imbens, 2006; Bertrand et al., 2004).

To get naive OLS estimates of returns to education, I use Model 1:

$$Y_{ipry} = \beta_0 + \beta_1 Edu_{ipr} + X_{ipr} + \lambda_p + \eta_r + \mu_y + \varepsilon_{ipry} \tag{1}$$

where Y_{ipr} is the labor market outcome of graduate i from program p in region r in year y (I look at log monthly wages and a dummy variable equal to 1 if a graduate is employed and 0 otherwise), where Edu_{ipr} is the imputed number of years of education of graduate i from program p in region r, X_{ipr} are exogenous individual characteristics of graduate i from program p in region r (sex, type of institution (public/private), and funding (state–funded/fee–based)), λ_p are program fixed effects, η_r are region fixed effects, μ_y are the year of graduation fixed effects.

I conduct a DiD analysis to estimate the effect of one year of higher education on labor market outcomes. The results of a DiD analysis are obtained from Model 2:

$$Y_{ipry} = \gamma_0 + \gamma_1 Treated_p + \gamma_2 Treated_p * Post_y + X_{ipr} + \lambda_p + \mu_y + \eta_r + \varepsilon_{ipry} \tag{2}$$

where Y_{ipr} is the labor market outcome of graduate i from program p in region r, $Treated_p$ is a dummy variable equal to 1 if a student has graduated from a 4–year program p and 0 otherwise; $Treated_p * Post_y$ is the interaction of $Treated_p$ and a dummy variable for the year of graduation 2015. γ_2 is the parameter of interest that gives the causal estimate of the effect of a one–year reduction in higher education. Only γ_2 is reported in the tables unless stated otherwise. To estimate the probability of being employed, I use a linear probability model. All results reported in this paper are obtained using sample weights. I cluster all standard errors at the program and the graduation year level.

4 Results

4.1 Main results

In this section, I discuss OLS estimates of returns to higher education and present the results of the DiD estimates of a one-year reduction of higher education on wages and employment.

OLS estimates of returns to higher education

Table 3 shows the OLS estimates as baseline results of returns to one year of higher education in Russia. Columns (1), (2), and (3) show the association between one year of higher education and log monthly wages for all graduates, and separately for males and females. Columns (4), (5), and (6) indicate the results of the linear probability model for employment. The estimated associations can be interpreted analogously to returns to one year of higher education. The results for log monthly wages show that returns to one year of higher education are 4.5%. Returns are higher for females than for males, which is consistent with the current literature. The overall results are mostly driven by returns to higher education for females. Returns to one year of higher education are 5.7% for females, while returns to education for males are 3.5%. All the estimates are statistically significant. The results for employment indicate that one year of education is associated with an increase in the probability of being employed by 2.3 percentage points. The results are driven mostly by female graduates: one year of education is associated with a 2.7 percentage points increase in the employment probability, while the effect for males is not statistically significant. Nonetheless, the OLS results are most likely biased upwards, as endogeneity bias arises due to people with higher marginal returns to education choosing a higher level of schooling. Thus, I estimate the causal effect of one year of higher education, using a DiD design.

The causal effect of the Bologna reform on wages

Table 4 shows the results for log monthly wages and employment, using a DiD design. The main variable is *Treated*Post*, which shows the causal effect of a one–year reduction in higher education. Columns (1) and (2) yield the results for log monthly wages. I start with a minimal specification in Column (1) that includes only year and region fixed effects. In Column (2) I add individual controls, including sex, type of institution (public/private), and funding (state–funded/fee–based). The parsimonious specification in Column (1) is the preferred one. Column (1) shows that returns to one year of higher education are not statistically significant.

The previous findings use log monthly wages to estimate the effect of one year of higher education. One might expect that individuals with more years of education

Table 3: OLS estimates of the association between years of higher education and log monthly wages and employment

	Log monthly wages			*Employed*		
	All	Male	Female	All	Male	Female
	(1)	(2)	(3)	(4)	(5)	(6)
Years of education	0.045**	0.035*	0.057**	0.023**	0.015	0.027**
	(0.013)	(0.014)	(0.017)	(0.007)	(0.010)	(0.011)
Pre–reform mean	10.02	10.02	10.02	0.88	0.88	0.88
Observations	6,207	2,771	3,436	11,332	4,783	6,549

Notes: Robust standard errors clustered at the program and the graduation year level in parentheses. The dependent variable is imputed years of education. All columns include year and region fixed effects. Columns (1) and (4) include individual controls for sex, type of institution (public/private), and funding (state–funded/fee–based). Columns (2), (3), (5), and (6) include the same individual controls as columns (1) and (4), excluding sex.

*p<0.1; **p<0.05; ***p<0.01

receive higher monthly wages because they work more hours, or because their productivity increases so their hourly wages increase. Thus, I estimate Model 2, using log hourly wages as a dependent variable. Column (3) of Table 4 yields identical results, suggesting that a one–year reduction does not cause changes in the working hours or productivity of graduates. All in all, the results suggest that the OLS estimates of log monthly wages in Table 3 are upward biased, and returns to one year of higher education are close to zero and not significantly different from it.

The causal effect of the Bologna reform on employment

Similarly, I estimate the effect of a one–year reduction in higher education on employment. Column (5) of Table 4 includes only year and region fixed effects, Column (6) adds individual controls, and the parsimonious model is the preferred one. Column (5) shows that a one–year reduction in higher education does not have a statistically significant effect on employment. Overall, the results confirm that a one–year reduction does not affect the labor market outcomes of university graduates.

Table 4: DiD estimates of the effect of a one–year reduction of higher education on log monthly wages, hourly wages, and employment for all

	Log monthly wages		*Log hourly wages*		*Employed*	
	(1)	(2)	(3)	(4)	(5)	(6)
Treated*Post	−0.010	−0.018	−0.021	−0.026	−0.014	−0.015*
	(0.015)	(0.012)	(0.016)	(0.022)	(0.007)	(0.007)
Ind. controls		✓		✓		✓
Pre–reform mean	10.02	10.02	4.91	4.91	0.88	0.88
Observations	6,207	6,207	6,207	6,207	11,332	11,332

Notes: Robust standard errors clustered at the program and the graduation year level in parentheses. Columns (2), (4), and (6) include individual controls for sex, type of institution (public/private), and funding (state–funded/fee–based).

*p<0.1; **p<0.05; ***p<0.01

4.2 Heterogeneity results

To examine whether the effects of a one–year reduction on wages and employment are heterogeneous across different groups, I estimate the model separately by sex and ability.

By sex

Columns (1) and (2) of Table 6 in the appendix show the results of a reduction of higher education on log monthly wages separately for males and females. I use the parsimonious model and include only year and region fixed effects. There is a negative statistically significant effect of a one–year reduction for male graduates while the effect is not statistically significant for female graduates. The estimates for female graduates allow to rule out effects larger than 3%, which is half in size compared to OLS estimates.

Columns (3) and (4) of Table 6 in the appendix show the results of a reduction of higher education on employment separately for males and females. The model specification is the same as in Columns (1) and (2). In line with previous results for the pooled sample, a one-year reduction does not induce a reduction in the probability of being employed for male and female graduates. Thus, there are no significant sex differences in the effect of a reduction in the length of higher education on employment.

By ability

As a proxy for the ability I use a dummy variable for whether a graduate studied at a state–funded or fee–based place. The reason to use this dummy is that applicants with higher scores from the standardized exam for admission are more likely to study at a state-funded place at university, thus, this can be used as a proxy for ability. The results are presented in Table 7. The model specifications are the same as in Table 6. The effect is not statistically significant from zero for wages and employment both for students who study at state–funded places and at fee–based places. This indicates that people with different ability levels do not respond differently to the reform.

4.3 Mechanism

What is a potential explanation for the heterogeneous effects of the reform by sex? In this section, I provide suggestive evidence about why male and female graduates have been affected differently. I argue that a possible explanation for these results is different strategies chosen by male and female students to compensate for losses in their human capital, i.e. the compensation–based mechanism.

One might argue that 5–year degrees could have been compressed into 4–year programs without losing much of the human capital because students could study more outside of the classroom given enough study materials. Although I do not have information on the number of hours graduates spend studying, I can use a proxy for the amount of time students spend on non–studying related activities, for example, whether students work while studying. If 4–year students would have to spend more hours studying compared to 5–year students, they would have less time to work whilst studying.

Column (1) of Table 5 yields a positive effect of the reform on whether a male student worked while studying, while Column (2) shows that there is a negative effect for females. These results provide a potential explanation for why males and females have been affected differently by the reform. Female students decided not to work during their shortened university program to compensate for human capital losses, while male students chose to work more. These findings indicate that male students considered working while their studies as a form of investing in

their human capital to compensate for their losses, while female students considered working while studying as a harmful practice for their level of human capital.

To support this claim, one would need to have a proxy for the level of accumulated human capital. Since male students worked more during their studies, while female students worked less, one can expect that their accumulated human capital would differ. As human capital theory suggests, shortening the length of the university study period reduces the direct and indirect costs of obtaining a degree (Schultz, 1961; Becker, 1962). The reduction in the length of study leads to the decreased productivity of a graduate, which will result in lower wages and unfavorable labor market opportunities. However, some students can compensate for the reduction in the duration of their studies and invest in the accumulation of human capital by studying more outside of the classroom.

There are some questions in the dataset that help to support this claim. The most relevant question for the paper is whether graduates had to get on-the-job training due to insufficient competence. About 14% of 4–year and 5–year graduates said that they got on–the–job training due to insufficient competence. Column (3) of Table 5 shows that there is a positive effect of the reform on the probability of getting on-the-job training for males, while there is a negative effect for female graduates. The estimated coefficients are quite sizable on their own, but especially when they are compared to the pre–reform means.

These results suggest that male graduates decreased their accumulated amount of human capital, while females did not. As a result, male graduates had to take more on–the–job training than females did. By combining the results for the time spent outside of the classroom and for the accumulated level of human capital, I conclude that female students could compensate for the reduction in the duration of higher education by working less during their studies, such that they did not decrease their level of human capital. Male graduates worked more during their studies, and, as a result, they could not compensate for the losses in their human capital and have been negatively affected by the reform.

Table 5: DiD estimates of the effect of a one–year reduction of higher education on combining study and work and the level of on-the-job training

	Combining study and work		*On-the-job training*	
	Male	Female	Male	Female
	(1)	(2)	(3)	(4)
Treated*Post	0.081***	−0.042**	0.072**	−0.045**
	(0.015)	(0.012)	(0.024)	(0.013)
Pre–reform mean	0.28	0.20	0.14	0.13
Observations	4,783	6,549	1,385	1,747

Notes: Robust standard errors clustered at the program and the graduation year level in parentheses. All columns include year and region fixed effects.

*p<0.1; **p<0.05; ***p<0.01

4.4 Robustness checks

In this section, I consider three possible threats to the identification strategy: changes in the cohort composition, an introduction of the standardized exam for admission to universities, and the 2015 Russian financial crisis.

Changes in the cohort composition

One might argue that school graduates could defer the year of application from 2010 to 2011 to start a 4–year instead of a 5–year program. However, 90% of school graduates apply to full–time university programs in the same year as they graduate from school in the observed period. Thus, there is no evidence for the Bologna anticipation effect as it is unlikely that university applicants could affect the year of their application.

One might expect that the reduction in the length of the university study period could attract more students, as earning a degree has reduced the direct and indirect costs. In other words, this reform could have attracted more students into universities, who would not go to university otherwise. Given the shortening of the university study period, lower–ability individuals should be motivated to enroll in these programs. This might potentially create a bias, if more students were admitted to universities in 2011 and later on, compared to previous years. Figure 2 in the appendix shows the total inflow of 4– and 5–year full–time students into public

and private universities in 2005–2019. The total number of admitted students is decreasing over time, due to demographic changes. As evident in Figure 2, there is no increased inflow of students in the year of the reform and the following years, suggesting that the reform did not change the pool of admitted students.

Table 8 in the appendix shows the descriptive statistics for university students by the year of their graduation. The share of male students, the share of students from public universities, and the share of students who have state–funded places is consistent throughout all years. Overall, the observed characteristics of the cohorts of students are stable over time.

Ultimately, the first cohort of students, who were admitted to 4–year programs in 2011, graduated in 2015, and the last cohort of students, who were admitted to 5–year degrees in 2010, also graduated in 2015. Thus, the proportion of 4–year graduates to 5–year graduates was 1:1 in 2015 which confirms that there was an equal number of university applicants in 2010 and 2011 who could finish their studies on time (see Figure 3 in the appendix). This indicates that the implementation of the reform has not led to a shift in the composition of graduates.

An introduction of the standardised exam for admission to universities

The standardized exam for admission to universities – the Unified State Exam (similar to the SAT) – was implemented in some regions in 2003 and was introduced nationwide in 2009. Thus, it increased the accessibility of universities for students with different backgrounds. Table 9 in the appendix shows the observable characteristics of graduates when they were admitted to a university. There are no differences in the age of students at which they were admitted to university: the mean age is 17.6. The share of students from rural areas is slightly higher in the period 2013–2015 than in the period 2010–2012. The share of students from the most developed cities, Moscow and Saint Petersburg, remained stable over the period. Thus, this reform does not influence the results of the paper as students were equally affected by this reform.

The 2015 Russian financial crisis

Another confounding factor is that there was a crisis in Russia in 2015. The literature shows that students, who graduate in a recession, suffer lower wages and experience unfavorable labor market conditions (Oreopoulos et al., 2012; Kahn, 2010). Russian graduates could have suffered due to the crisis in 2015, which is the year of interest. To check the robustness of my findings, I use data from the same source for students who graduated in 2010–2018.[6] The drawbacks of this dataset are that there is no information on whether a student graduated from a part– or full–time program, and there is no information on wages. Although there might arise a bias, due to the inclusion of part–time graduates, I expect this bias to be consistent for the treatment and control groups. I estimate the following even-study specification:

$$Y_{ipr} = \alpha_0 + \sum_{\tau=-q}^{-1} \alpha_\tau D_{p\tau} + \sum_{\tau=0}^{m} \delta_\tau D_{\tau p} + X_{ipr} + \lambda_p + \mu_y + \eta_r + \varepsilon_{ipr} \tag{3}$$

where $D_{p\tau}$ is a dummy variable equal to 1 if a student has graduated from a 4–year program p in year τ and 0 otherwise. I include q leads and m lags, and the reform occurs in year 0.

In total there are 4 cohorts of university students who graduated after the reform (2015–2018). The estimates of the effect of the reform on employment in the even-study framework are presented in Figure 4 in the appendix. Although the magnitude of the effect seems to be increasing over time, there is no evidence to reject the null effect. These results support a zero effect of a one–year reduction of higher education on employment for graduates in 2015–2018.

Figure 4 also provides the estimates of the pre-treatment leads with 95% confidence intervals, i.e. the estimates of the interaction of the treatment dummy and pre–reform period dummies. Ultimately, this test of the parallel pre–trends seems to support the parallel trend assumption, suggesting that before 2015 labor market conditions for the treated and control groups followed similar patterns (Cunning-

[6]The data are obtained from the Russian Labor Force Survey carried out by the Russian Federal State Statistics Service (Rosstat). The restrictions described in Section 3 are applied to this dataset.

ham, 2021).

5 Concluding remarks

The main finding of the paper is that the Bologna reform did not harm the employment of graduates. Moreover, the wages of students who invested more time into studying have not been affected either. I show that the DiD estimates are smaller than the OLS estimates of returns to higher education which is intuitive as a positive relationship between education and unobserved ability is expected which creates an upward bias of the naive OLS estimates.

As forgone earnings and tuition fees are the real costs of going to university instead of working, 4–year graduates have lower costs for earning a higher education degree. This suggests that the reform decreased the university study period, forgone earnings, and tuition fees without negatively affecting labor market outcomes for graduates who chose to work less and study more. The reform has shortened the university study period, allowing 4–year graduates to have an additional year of labor market experience compared to those who completed 5–year programs. As the literature suggests, additional work experience has an impact on lifetime earnings, meaning the reform has increased the lifetime wealth of 4–year graduates (Dustmann and Meghir, 2005; Topel, 1991).

Ultimately, the results shed light on the causal effects of higher education on the labor market in middle–income countries. A limitation of this paper is that only short–term effects of a one–year reduction of higher education are estimated. Further research can use the Bologna reform as a source of an exogenous variation in the length of education and estimate long–term labor market effects. The reform can be used to study other effects of higher education on, for example, lifetime wealth and health, as it offers a rare exogenous variation for producing convincing estimates.

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6 Appendix

Table 6: DiD estimates of the effect of a one–year reduction of higher education on log monthly wages and employment separately by sex

	Log monthly wages		*Employed*	
	Male	Female	Male	Female
	(1)	(2)	(3)	(4)
Treated*Post	−0.046**	0.017	−0.020	−0.012
	(0.015)	(0.015)	(0.032)	(0.022)
Pre–reform mean	10.13	9.92	0.95	0.82
Observations	2,771	3,436	4,783	6,549

Notes: Robust standard errors clustered at the program and the graduation year level in parentheses. All columns include year and region fixed effects.

*p<0.1; **p<0.05; ***p<0.01

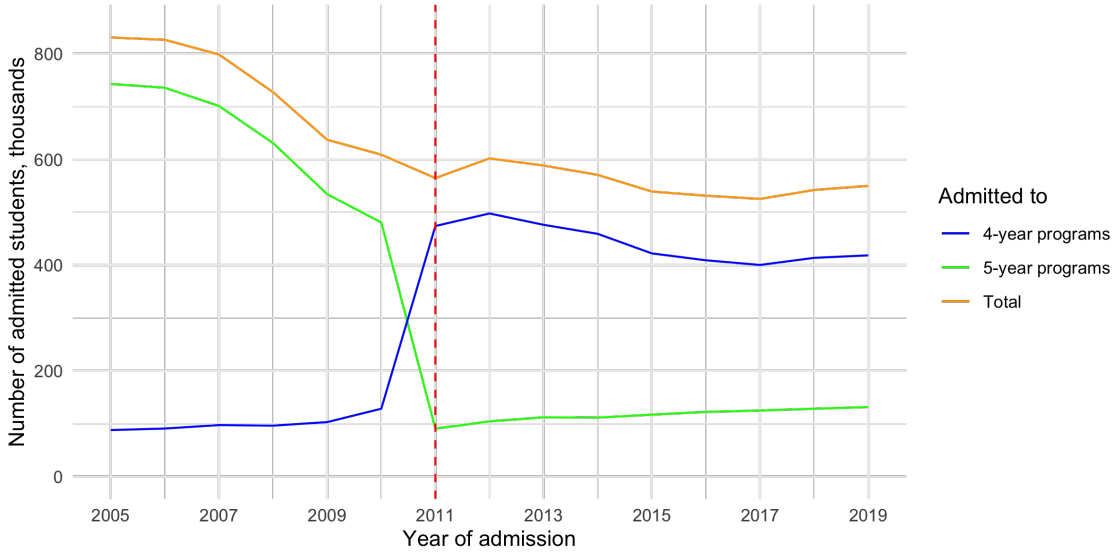
Table 7: DiD estimates of the effect of a one–year reduction of higher education on log monthly wages and employment separately by funding

	Log monthly wages		*Employed*	
	State–funded	Fee–based	State–funded	Fee–based
	(1)	(2)	(3)	(4)
Treated*Post	−0.013	−0.001	−0.001	−0.021
	(0.020)	(0.010)	(0.015)	(0.015)
Pre–reform mean	10.02	10.01	0.88	0.86
Observations	3,730	2,477	6,697	4,635

Notes: Robust standard errors clustered at the program and the graduation year level in parentheses. All columns include year and region fixed effects.

*p<0.1; **p<0.05; ***p<0.01

Figure 2: The number of 4– and 5–year full–time admitted students in public and private universities in 2005–2019, thousands



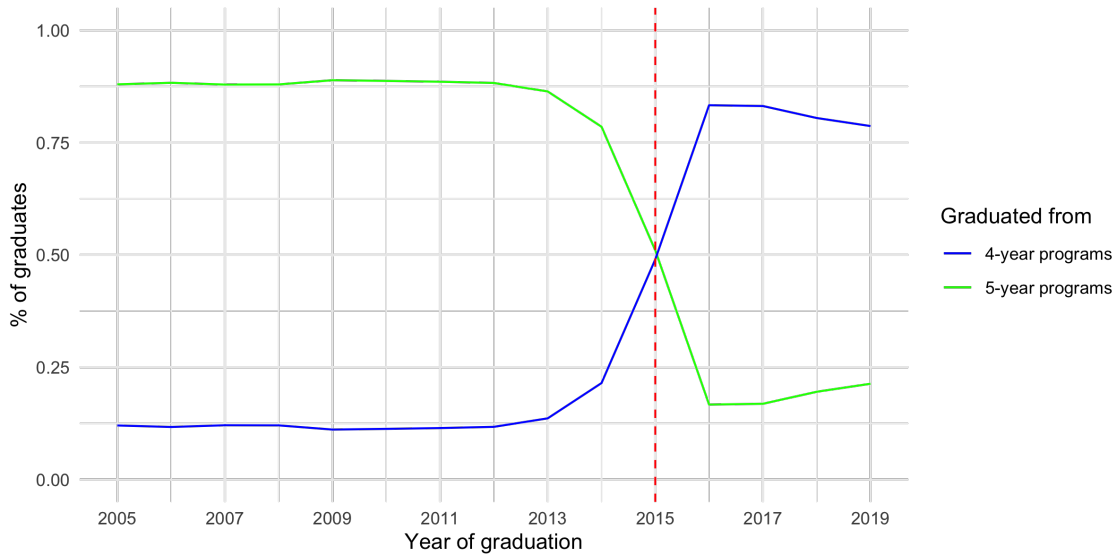
Notes: The red dashed line shows the year of the implementation of the reform.
Source: Ministry of Science and Higher Education.

Table 8: The characteristics of students by the year of graduation

Variable	2010	2011	2012	2013	2014	2015
Male	0.44	0.44	0.45	0.42	0.40	0.39
	(0.50)	(0.50)	(0.50)	(0.49)	(0.49)	(0.49)
Public institution	0.94	0.96	0.95	0.95	0.96	0.95
	(0.23)	(0.20)	(0.22)	(0.22)	(0.19)	(0.21)
State–funded	0.59	0.60	0.59	0.59	0.59	0.58
	(0.49)	(0.49)	(0.49)	(0.49)	(0.49)	(0.49)

Notes: This table shows sample means and standard deviations from the the National Survey of Graduate Employment carried out by the Russian Federal State Statistics Service.

Figure 3: The share of 4– and 5–year full–time graduates in public and private universities in 2005–2019



Notes: The red dashed line shows the year of a double flow of graduates.

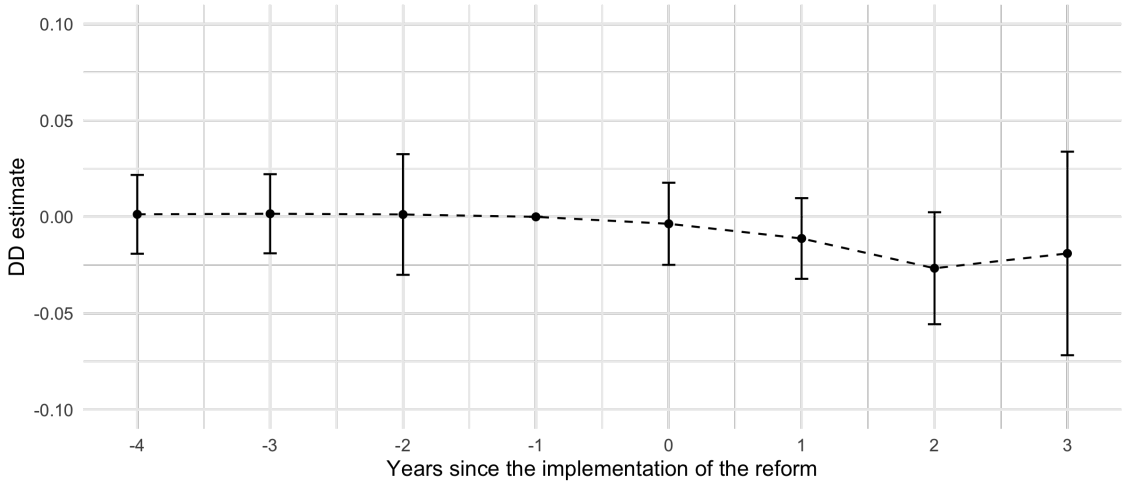
Source: Ministry of Science and Higher Education.

Table 9: The characteristics of graduates when they were admitted to universities

Variable	2010	2011	2012	2013	2014	2015
Age at the year of application	17.38	17.45	17.62	17.66	17.82	17.73
	(1.09)	(1.21)	(1.26)	(1.27)	(1.39)	(1.35)
Rural residence	0.21	0.23	0.24	0.28	0.27	0.28
	(0.41)	(0.42)	(0.43)	(0.45)	(0.44)	(0.45)
Moscow and Saint Petersburg	0.13	0.13	0.13	0.12	0.10	0.12
residence	(0.34)	(0.33)	(0.34)	(0.32)	(0.30)	(0.33)

Notes: This table shows sample means and standard deviations from the the National Survey of Graduate Employment carried out by the Russian Federal State Statistics Service.

Figure 4: Event-study estimates of a one–year reduction of higher education on employment



Notes: The figure shows the regression coefficients of interaction between the treatment variable and time period dummies with 95% confidence intervals. The omitted period is -1, the year before the implementation of the reform.